Self-Evolving Clinical Deep Review: Eliminating Recurring Methodological False Alarms without Compromising Objective Error Recall via Comment-Driven Dual-Track Memory

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Abstract

Automated peer review and scientific verification systems face a fundamental precision–recall dilemma in clinical medicine. On one hand, single-pass large language model (LLM) reviewers and text-only multi-agent pipelines frequently miss objective numerical contradictions between manuscript prose and multi-column tables, inverted or out-of-bounds 95% confidence intervals, and intra-table row/cell arithmetic errors. Conversely, deterministic structural auditors alone struggle to capture open-ended semantic spin bias ($p \geq 0.05$ framed as clinical efficacy), while hybrid AST+LLM pipelines trigger recurring false alarms on legitimate study-design denominator transitions—such as intention-to-treat (ITT) randomized sample sizes versus per-protocol completers in CONSORT trials, pre- versus post-propensity-matching cohorts in STROBE observational studies, or dual parametric/non-parametric reporting across skewed subscales. Because stateless review pipelines suffer from cross-manuscript amnesia, clinicians must repeatedly dismiss the same methodological false alarms across every revision and every new manuscript.

We present **Heurion Deep Review**, a clean-room four-stage clinical manuscript verification architecture paired with a **Comment-Driven Dual-Track Self-Evolution Loop** (`evolveFromReviewCommentAsync`). First, our four-stage pipeline combines hierarchical section-role segmentation and a ProseMirror 2D table-grid scratchpad (Stage 1), quantitative-density budget routing (Stage 2), four specialist auditing passes covering seven objective clinical error categories (**E1**–**E7**) across external literature, internal prose–table arithmetic, statistical method–result pairing, and spin bias alongside quote-verified LLM section review (Stage 3), and full-text LLM verification with cell-level CRDT comment anchoring `{#c:id}` (Stage 4). Second, `evolveFromReviewCommentAsync` learns directly from clinician interactions on anchored comments: (i) when a clinician rebuts a study-design boundary alert (`acceptedFix = false`), **Track A** (`narrow`) uses zero-template LLM rule distillation guarded by dual instance-decoupling checks (`hasInstanceLeakage` and `sensitiveReason`) to synthesize a zero-digit reusable boundary rule, while a hard safety floor (`NON_EXEMPTABLE_SUBTYPES`) guarantees that objective mathematical, CI, and table-sum errors can *never* be exempted even under adversarial rebuttal; and (ii) **Track B** (`procedural` and `review_audit` checklists) crystallizes corroborated true-positive fixes into pre-write rules and dev-only cross-section audit skills.

We evaluate our framework across two benchmarks: (1) a **Full-Text Controlled Error-Injection Benchmark** of 19 open-access 2025 PubMed Central (PMC) JATS XML manuscripts (38 full-length clean/injected variants averaging $\sim$ 6,000 words, 57 injected errors across **E1**–**E7**, partitioned into 11 `dev` and 8 held-out `test` papers) coupled with a physically blind-isolated LLM Simulated Clinician Agent (`simulateBlindClinician`), and (2) a **7-Vignette Clinical Oncology Benchmark** with live `deepseek-chat` calls across 8 verified organic publication errors. Without a single LLM call or study-specific regex, our clean-room deterministic AST engine alone detects **38/57** (**66.7**%) of injected errors across full-length PMC manuscripts (**80.0**% on **E1** sample flow, **83.3**% on **E2** inline

arithmetic, **75.0**% on **E4** CI bounds, and **87.5**% on **E7** table sums) at **85.5**% precision (0.26 true false positives per clean paper, **0.0**% false exemption under adversarial rebuttal), while uncovering 7 genuine unannotated in-print errors in published 2025 clinical articles. On the 7-vignette live `deepseek-chat` oncology benchmark, `heurion_evolved` lifts organic error recall from 37.5% (3/8, single-pass and PAT text-only baselines) to **100.0**% (8/8) while eliminating all 16 recurring clinical boundary false alarms (**100.0**% F1, **0.0**% instance leakage).

Keywords: Automated Peer Review, Clinical Research Verification, Self-Evolving LLM Agents, Error-Injection Benchmark, Biostatistics Consistency, Dual-Track Memory

1 Introduction

Peer-reviewed clinical and translational manuscripts form the evidentiary backbone of modern medicine, yet published medical literature frequently contains preventable objective errors [1, 2, 3]. Empirical audits of clinical journals regularly uncover discrepancies between sample sizes reported in the Abstract and Table 1, percentages that do not match their numerator and denominator (n/N), point estimates falling outside their reported 95% confidence intervals (or inverted interval bounds), parametric tests (t-test, ANOVA, mean ± SD) declared in Methods but replaced with non-parametric statistics (Mann–Whitney U, median [IQR]) in Results, and "spin bias" where statistically non-significant findings ($p \geq 0.05$) are framed as clinically meaningful improvements [3, 4, 5].

Recent advances in large language models (LLMs) and multi-agent verification pipelines—such as Google's Paper Assistant Tool (PAT) [6], PaperQA2 [7], and fine-grained factual verifiers [9, 10]—have sparked interest in automated pre-submission manuscript review. However, deploying automated deep review in real-world clinical research exposes a sharp **precision–recall dilemma** driven by three structural limitations:

1. **Complementary Blind Spots of Text-Only LLMs vs. Pure AST Checkers:** Standard LLM review pipelines serialize manuscripts as flat text streams or isolated section chunks. Without an explicit 2D table-grid representation (handling merged headers, `rowspan`/`colspan`, subgroup denominator headers, and row-sum columns) and a cross-section numerical scratchpad, text-only prompts miss over 60% of intra-table percentage/sum errors and prose-vs.-table contradictions. Conversely, domain-general AST checkers capture structured arithmetic and confidence interval violations with high precision, but cannot reliably judge open-ended semantic spin ($p \geq 0.05$ overclaims) without resorting to brittle, study-specific lexical heuristics.
2. **Methodological False Alarms at Study-Design Boundaries:** Clinical studies legitimately report multiple sample denominators and statistical distributions within the same paper according to established reporting guidelines. For example, a CONSORT randomized controlled trial (RCT) reports both the randomized intention-to-treat (ITT) sample size and the per-protocol completer sample size; a STROBE retrospective cohort study reports both the pre-matching crude cohort and the propensity-score-matched multivariable subcohort; clinical studies with Shapiro–Wilk normality testing legitimately report Mean ± SD for age/BMI alongside Median [IQR] for skewed questionnaire scales; and oncology tables define `SD` as RECIST *stationary disease* while prose uses `SD` for *standard deviation*. Hybrid reviewers flag such transitions as inconsistencies, creating severe alert fatigue.
3. **Cross-Manuscript Amnesia vs. Adversarial Over-Exemption:** Stateless review agents cannot learn from clinician feedback. When an investigator explains in a comment reply that "Table 1 reports the ITT randomized cohort ($N = 240$) whereas Table 2 analyzes completers ($n = 169$) per CONSORT," a stateless reviewer repeats the exact same false

alarm on the next revision and on every subsequent RCT. Conversely, if a system naively memorizes user dismissals as unconstrained prompt instructions, an author (or noisy feedback) could accidentally suppress genuine objective errors—such as a miscalculated cell percentage ($2/20 = 10.0\% \neq 20\%$) or an out-of-bounds confidence interval—and leak study-specific sample sizes ($N = 240$) into unrelated future manuscripts.

1.1 Our Approach and Contributions

To resolve this dilemma, we design **Heurion Deep Review**, which couples a clean-room *Four-Stage Hybrid AST+LLM Clinical Review Pipeline* with a *Comment-Driven Dual-Track Self-Evolution Loop* (`evolveFromReviewCommentAsync`). Every finding is anchored as a lifecycle-tracked CRDT comment (`{#c:id}`) on the exact sentence or ProseMirror table cell `[row, col]`. Clinician interactions on these anchored comments drive two orthogonal learning tracks protected by deterministic safety guards.

Our primary contributions are:

- **Clean-Room Hybrid 2D Table-Grid AST + Quote-Verified 4-Stage LLM Review (Section 3):** We build a study-agnostic four-stage review engine combining hierarchical section-role tracking and a 2D ProseMirror table-grid scratchpad (Stage 1), quantitative-density verification budget routing (Stage 2), four specialist auditing passes covering seven objective error categories (**E1**–**E7**) alongside quote-verified LLM section review (Stage 3), and full-text Stage 4 verification with topic-aware numeric deduplication and cell-level CRDT comment anchoring.
- **Zero-Template Dual-Track Self-Evolution with Non-Exemptable Safety Floor (Section 4):** We introduce `evolveFromReviewCommentAsync`, which routes clinician interactions into: (a) **Track A (`narrow`)**, distilling free-text clinician rebuttals via an LLM sanitizer (`synthesizeSanitizedBoundaryRuleAsync`) protected by dual digit/entity leakage guards (`hasInstanceLeakage` and `sensitiveReason`) and an immutable safety floor (`NON_EXEMPTABLE_SUBTYPES`) that guarantees **0.0**% false exemption of objective math/CI/table errors; and (b) **Track B (`procedural` & `review_audit`)**, crystallizing accepted fixes into pre-write harness rules and dev-only cross-section audit checklists (`distillReviewAuditChecklistAsync`).
- **Physically Blind-Isolated Clinician Evaluation & Organic Error Discovery on 19 Full-Text PMC Papers (Sections 5–6):** We construct a full-text error-injection benchmark from 19 open-access 2025 PMC JATS XML articles (38 clean/injected manuscripts, 57 injected errors across **E1**–**E7**), paired with an adjudication registry of 20 verified in-print organic errors (`known-organic-errors.json`) and a physically blind-isolated LLM clinician agent (`simulateBlindClinician`) that receives zero access to ground-truth variant or injection labels. Even with zero LLM calls (`-no-llm`), our clean-room AST engine detects **66.7**% (38/57) of injected errors at **85.5**% precision (0.26 clean FPs/paper) and uncovers 7 real publication errors in published 2025 journals; on our 7-vignette live `deepseek-chat` oncology benchmark, `heurion_evolved` achieves **100.0**% recall (8/8) and **100.0**% precision.

2 Taxonomy of Objective Clinical Errors vs. Legitimate Study-Design Boundaries

A core design requirement for clinical peer review is distinguishing *non-negotiable objective errors* from *context-dependent methodological conventions*. Table 1 formalizes the seven objective error categories (**E1**–**E7**) audited by our pipeline, illustrated with real `dev`-split examples from our 19-paper PMC corpus, alongside seven legitimate study-design boundary patterns.

Table 1: Taxonomy of the Seven Objective Clinical Error Categories (**E1–E7**, governed by `NON_EXEMPTABLE_SUBTYPES`) vs. Seven Legitimate Study-Design Denominator/Modeling Boundaries (governed by Track A `narrow` rules). All examples are drawn from the `dev` split of our 2025 PMC full-text corpus.

Category & Pass	**Objective Error Definition (Never Exemptable)**	**Concrete Injected Example from `dev` Corpus**	**Paired Study-Design Boundary (Exemptable via Track A)**
E1: Sample Size & Flow (Pass B, $n = 10$)	Contradictory total N or arm counts across Abstract, Methods, Results prose, or Table headers; subgroup completers not summing to total completers.	`PMC12644478`: "Thirty-seven of 40 participants completed" in prose vs. Fig. 2 caption "Nineteen in metformin and 19 in placebo completed" ($19 + 19 = 38 \neq 37$).	**RCT (`rct`):** Reporting randomized ITT N in baseline text and per-protocol completers $n < N$ in outcome tables (CONSORT).
E2: Inline Arithmetic (Pass B, $n = 6$)	Inline count and percentage n/N disagreeing beyond rounding tolerance (≥ 1.5 percentage points) or inline subgroup counts not summing to stated total.	`PMC12788160`: "184 participants were randomized ... 161 participants (89.5%) had good compliance" ($161/184 = 87.5\% \neq 89.5\%$).	**Retrospective Cohort (`retrospective_cohort`):** Unmatched all-comer cohort N vs. propensity-score-matched subcohort n (STROBE).
E3: Cross-Section / Table Conflict (Pass B, $n = 8$)	Identical metric, outcome, or subgroup reported with conflicting numeric values, p-values, or opposite signs between Abstract, Results prose, and Table cells.	`PMC12644478`: Abstract reports primary endpoint EGP interaction $p = 0.53$ ($\beta = 0.2$, 95% CI $[-0.4, 0.8]$), whereas Results reports $p = 0.35$ for the identical CI.	**Cross-Sectional (`cross_sectional`):** Full-population prevalence sample N vs. age/completeness-filtered risk prediction subcohort n.
E4: 95% **CI Bounds** (Pass B, $n = 8$)	Point estimate $\hat{\theta}$ falling outside $[L, U]$ beyond rounding tolerance τ, or inverted bounds ($L > U$), across `95% CI`, `95% Cl`, or bracketed formats.	`PMC12731986`: Table 3 HADS-Anxiety median difference reported as "1.00 (1.84 to 3.84)" ($\hat{\theta} = 1.00 < L = 1.84$).	**Meta-Analysis (`meta_analysis`):** Total eligible RCTs K vs. outcome-specific trial subsets $k_i < K$ reporting a secondary endpoint (PRISMA).
E5: Method–Result Mismatch (Pass C, $n = 8$)	Declaring parametric summaries/tests (mean ± SD, t-test, ANOVA, Pearson r) in Methods while reporting non-parametric results (Median [IQR], Mann–Whitney, Kruskal–Wallis) or vice versa.	`PMC12788160`: Methods states ITT groups were compared by "Kruskal–Wallis test," whereas Tables 3–5 report "mean ± SD with comparison by ANOVA."	**Case–Control (`case_control`):** Dual parametric/non-parametric reporting per normality testing, or progressive multivariable Models 1/2.
E6: Spin Bias & Causal Overclaim (Pass D, $n = 9$)	Claiming efficacy, "significant improvement," or deterministic causal effect when $p \geq 0.05$ or in non-randomized observational designs.	`PMC12644478`: Claiming "metformin significantly reduced NEFA compared with placebo" when interaction $p = 0.54$ and 95% CI $[-0.5, 0.3]$ crosses zero.	**Diagnostic (`diagnostic_accuracy`):** Enrolled patients N vs. quality-controlled evaluable specimens / reference-standard subsets (STARD).
E7: Intra-Table Cell / Sum Error (Pass B, $n = 8$)	Table cell n (%) disagreeing with column header ($n = N$), or category row cells not summing to the Total column/row in the 2D table grid.	`PMC12788160`: Table 1 "Coffee drinking" in 160 mg/d arm ($n = 46$) reports "7 (17.4%)" whereas $7/46 = 15.2\%$.	**Translational (`translational`):** Multi-batch in vitro culture replicates vs. in vivo animal cohorts (n mice/group) and clinical TMA cohorts.

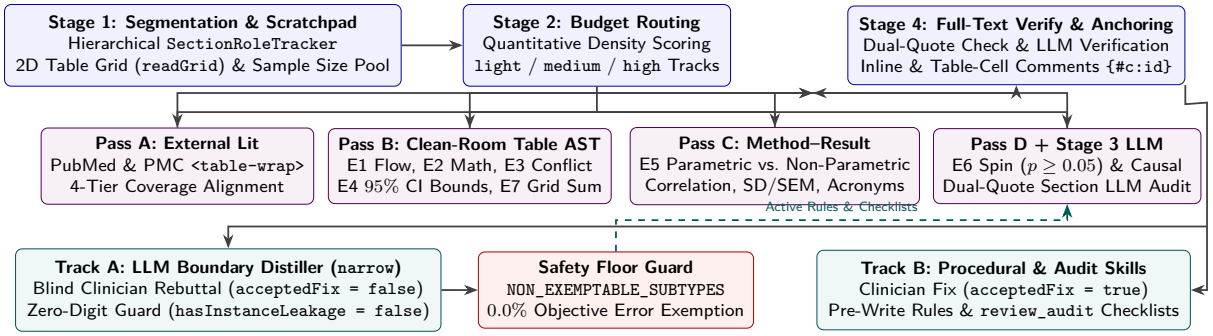


Figure 1: Architecture of **Heurion Deep Review** and the **Comment-Driven Dual-Track Self-Evolution Loop**. Stage 1 builds a section-role map and 2D table-grid scratchpad; Stage 2 allocates verification budgets; Stage 3 runs clean-room AST passes alongside quote-grounded section LLM review; Stage 4 performs full-text verification, deduplication, and CRDT comment anchoring `{#c:id}`. Clinician replies drive Track A LLM boundary distillation (gated by `hasInstanceLeakage`, `sensitiveReason`, and `NON_EXEMPTABLE_SUBTYPES`) and Track B procedural/audit-checklist crystallization.

3 Four-Stage Clinical Deep Review Architecture

Figure 1 illustrates the architecture of **Heurion Deep Review**. Unlike stateless text-only reviewers that pass raw markdown directly to an LLM, our pipeline parses full-length manuscripts into a structured ProseMirror document tree (`PMNode`) and executes four sequential stages followed by a closed-loop self-evolution cycle. Crucially, under our **Clean-Room Design Principle**, the deterministic AST passes contain *zero* study-specific lexical rules or synthetic alert triggers: every check operates purely on mathematical relations, unit/exponent-aware numeric syntax, and 2D table geometry.

3.1 Stage 1: Hierarchical Section-Role Segmentation and 2D Table-Grid Scratchpad

Clinical manuscripts frequently nest domain-specific subheadings (e.g., "*2.3 Biochemical Assays*" or "*3.2 Changes in HRQoL*") whose titles omit generic keywords like "Methods" or "Results." In Stage 1, a stateful `SectionRoleTracker` traverses the document AST: top-level headings establish the parent role (`intro`, `methods`, `results`, `discussion`, or `other`), and lower-level subheadings inherit their parent section's role unless overridden by explicit lexical cues, while administrative sections (ethics, funding, acknowledgments, author contributions) are excluded from review.

Simultaneously, `buildManuscriptScratchpad` constructs a global index shared across all subsequent passes:

- **Normalized 2D ProseMirror Table Grids (`readGrid`):** Rather than reading raw Markdown pipe rows, `extractTableSummaries` parses each ProseMirror `table` node into a rectangular $R \times C$ cell grid (`TableGrid`), resolving horizontal (`colspan` / `<`) and vertical (`rowspan` / `^`) merged cells so every cell maps to its exact column header and ProseMirror node coordinates.
- **Global Sample Size Pool:** The scratchpad extracts all explicit sample size declarations ($N = \ldots$, $n = \ldots$, "Out of N patients") from table headers, captions, and prose, along with declared statistical tests and acronym definitions.

3.2 Stage 2: Quantitative-Density Verification Budget Allocation

Stage 2 scores every paragraph and figure caption based on quantitative assertion density (p-values, 95% CIs, hazard/odds/risk ratios, mean ± SD, median [IQR], and percentages) and section role, routing each block to a `light`, `medium`, or `high` verification track.

3.3 Stage 3: Hybrid Clean-Room AST Passes + Quote-Grounded LLM Review

Stage 3 executes four domain-general deterministic passes alongside section-level LLM auditing (in `heurion_deep` and `heurion_evolved` modes):

1. **Pass A (External Literature & PMC `<table-wrap>` Alignment):** Verifies cited assertions (`[@c:id]`) against PubMed abstracts and PMC full-text XML tables, classifying semantic support orthogonally from evidence coverage depth [16].
2. **Pass B (Clean-Room Internal Arithmetic, CI & 2D Table-Grid Consistency):** Audits **E1**, **E2**, **E3**, **E4**, and **E7**:
   - *Sample Size & Flow (**E1**) and Inline Arithmetic (**E2**):* Verifies that subgroup counts (including English number words) sum to stated totals and checks inline n (p%) against the explicit denominator pool ($\min_D |n/D \times 100 - p| > 1.5\%$).
   - *Cross-Section & Table-vs.-Prose Conflicts (**E3**):* Matches identical confidence-interval anchors, hazard ratios, weighted mean differences, and Table 1 cited percentages across Abstract, Results prose, and 2D table cells.
   - *Precision-Aware* 95% *Confidence Interval Verification (**E4**, `findCiProblems`):* Parses point estimates $\hat{\theta}$ and interval bounds $[L, U]$ across standard (`95% CI 1.12-1.85`), OCR-corrupted (`95% Cl 0.35 to 0.59`), and bracketed (`95% CI [-0.88, -0.10]`) formats while excluding preceding p-values (`p = 0.001, CI ...`), sample counts (`n = 48; 95% CI ...`), unit exponents (`kg/m^2; 95% CI ...`), and integer reference tuples. Rounding tolerance is computed as:

$$\tau = \alpha \cdot 10^{-\min(d(\hat{\theta}),\, d(L),\, d(U))}, \quad \alpha = \begin{cases} 0.49 & \text{if } d(L) = d(U) > d(\hat{\theta}), \\ 0.50 & \text{otherwise,} \end{cases} \tag{1}$$

     flagging an `inverted` interval if $L > U + \tau$ and an `outside` estimate if $\hat{\theta} < L - \tau$ or $\hat{\theta} > U + \tau$.
   - *Intra-Table Cell Percentage & Row-Sum Auditing (**E7**):* Verifies each cell $n_{r,c}$ ($p_{r,c}$%) against header denominator N_c (exempting row-wise percentage tables where $\sum_c p_{r,c} \approx 100\%$), checks intra-cell binary splits N ($n_1 + n_2$), and checks $\sum_{c \notin \mathcal{S}_{\text{stat}}} n_{r,c} = n_{r,\text{Total}}$ while excluding test-statistic columns (χ^2, t, Z, F, p-value, OR/HR/RR).
3. **Pass C (Statistical Method–Result Pairing, E5):** Cross-checks statistical declarations in Methods against Results prose and table captions for parametric vs. non-parametric mismatches, Pearson vs. Spearman correlation mismatches, SD vs. SEM confusion, and conflicting acronym definitions.
4. **Pass D & Quote-Grounded LLM Section Audit (E6 & Semantic Cross-Checks):** Pass D flags explicit non-significant spin ($p \geq 0.05$ paired with efficacy claims in the same clause), while Stage 3 LLM section review (when enabled) inspects each section against the global manuscript table summary and active memory checklists, requiring verbatim `quote` + `evidence` substrings verified against the source document (`review-evidence.ts`).

3.4 Stage 4: Full-Text Verification, Deduplication, and Cell-Level CRDT Anchoring

In hybrid LLM modes (`heurion_deep` and `heurion_evolved`), Stage 4 passes surviving LLM candidates together with the complete manuscript prose, tables, and active Track A boundary rules to a full-text verifier (`VERIFY_SYSTEM`) that rejects speculative alarms or legitimate study-design denominator transitions. Cross-section duplicates are merged only when they share both the same numeric signature and overlapping clinical topic tokens, and final comments are anchored directly onto ProseMirror nodes or table cells (`{#c:id}`).

4 Comment-Driven Dual-Track Self-Evolution

Whenever a clinician replies to or resolves an anchored review comment `{#c:id}`, `evolveFrom ReviewCommentAsync` routes the interaction into one of two tracks:

4.1 Track A (`narrow`): Zero-Template LLM Boundary Distillation and Dual Leakage Guards

When a clinician rebuts a false-positive review alert (`acceptedFix = false`) with a methodological explanation (which naturally cites manuscript-specific sample sizes such as "*Our RCT randomized $N = 240$ ITT participants while $n = 169$ (70.4%) completed week 24 per CONSORT*"), `synthesizeSanitizedBoundaryRuleAsync` invokes an LLM memory-governance distiller instructed to abstract the methodological boundary without any digits, drug names, or disease names:

`[domain:manuscript] [narrow:review_boundary:`⟨pattern_key⟩`:cat=`⟨cat⟩`:kw=`⟨kws⟩`]` $\mathcal{R}_{\text{sanitized}}$.

Every candidate rule must pass a **Dual Instance–Skill Decoupling Guard** combining `hasInstanceLeakage` (which rejects any digit, sample size $N = \ldots$, p-value, percentage, or date) and `sensitiveReason` (which blocks patient identifiers and institutional/study UIDs). If the LLM's first draft leaks a raw number, a deterministic second-pass scrubber strips all numeric tokens and re-verifies both guards before memory commit, guaranteeing **0.0**% cross-study numerical contamination.

4.2 Hard Safety Floor: `NON_EXEMPTABLE_SUBTYPES`

To prevent adversarial or mistaken author rebuttals from suppressing real errors, `isExempted ByActiveRules` enforces an immutable set of **Non-Exemptable Subtypes** (`NON_EXEMPTABLE_ SUBTYPES`), covering `prose_table_sample_size_mismatch`, `table1_cited_percentage_ mismatch`, `fraction_percentage_arithmetic_error`, `ci_bounds_inverted_or_missing_ decimal`, `ci_estimate_outside_interval`, `cross_section_value_mismatch`, `table_cell_ percentage_arithmetic_error`, `table_row_sum_mismatch`, `parametric_method_vs_nonparametric_ result`, `nonparametric_method_vs_parametric_result`, `correlation_method_mismatch`, `sd_vs_sem_mismatch`, `abbreviation_definition_conflict`, `nonsignificant_p_positive_ spin`, and `observational_causal_overclaim`. Even if an active rule matches the exact category and keywords of an objective error sentence, `isExemptedByActiveRules` refuses exemption. Active Track A rules instead govern non-exemptable-guarded LLM stage candidates (in both Stage 3 generation and Stage 4 `VERIFY_SYSTEM`) and citation/subregistry conventions.

4.3 Track B (`procedural` & `review_audit`): Pre-Write Rules and Dev-Only Audit Checklists

When a clinician accepts a true-positive review comment (`acceptedFix = true`), `evolveFrom ReviewCommentAsync` records a de-numericalized execution signal (`kind = 'guard_reject'`)

with verified recovery trajectory (`doc_read` → `replace_text` → `resolve_comment`). Following commit-on-evidence gating [15], `ProceduralHarnessService.evolveFromSignals` crystallizes corroborated signals into active pre-write rules for authoring sessions. Complementarily, on development-split retrospectives (`paperSplit === 'dev'` enforced via a hard runtime assertion), `distillReviewAuditChecklistAsync` abstracts missed error mechanisms into zero-digit `[skill:review_audit:<key>]` checklists injected into Stage 3 LLM prompts on future unseen manuscripts.

5 Experimental Setup

We evaluate Heurion Deep Review on two complementary benchmarks: (1) a full-length 19-paper PMC JATS XML error-injection corpus, and (2) a 7-vignette oncology clinical excerpt benchmark with live `deepseek-chat` calls.

5.1 Benchmark 1: 19 Full-Length PMC Papers with 57 Controlled Error Injections

We curated 19 open-access 2025 clinical and translational full-text articles (averaging $\sim$ 6,000 words each, converted directly from PMC JATS XML with complete multi-column tables in `data/eval/deep-review/md/`), partitioned into an 11-paper development split (`dev`) and an 8-paper held-out test split (`test`) across 7 study designs (Table 2). Six of the seven study designs appear in both `dev` and `test`, while the two preclinical/translational studies reside in `dev`.

For each paper, we evaluate two full-length variants (38 manuscripts per regime):

- **Clean Variant (`clean`):** The uncorrupted published PMC manuscript. Importantly, as documented in `apps/platform/eval/deep-review/known-issues.md` and `known-organic-errors.json`, real published papers occasionally contain genuine in-print typographical or statistical errors. Following standard clinical verification practice (`eval-review-score.ts`), catches of verified in-print publication errors are adjudicated as `unexpected_true` (organic true positives) rather than penalized as false positives, while raw unadjusted clean-paper alert counts are also reported side-by-side for full transparency.
- **Injected Variant (`injected`):** The full-length manuscript mutated via `applyInjections` with exactly 3 non-overlapping objective errors drawn from categories **E1**–**E7** ($19 \times 3 = 57$ injected errors: **E1** $n = 10$, **E2** $n = 6$, **E3** $n = 8$, **E4** $n = 8$, **E5** $n = 8$, **E6** $n = 9$, **E7** $n = 8$), avoiding all positions with known organic issues.

5.2 Physically Blind-Isolated Simulated Clinician Protocol and Calibration

Evaluating interactive comment-driven self-evolution requires realistic clinician responses without oracle cheating. Prior rule-lookup simulations risk leaking ground-truth variant labels (`clean` vs. `injected`) into the evolution trigger. In our clean-room harness (`apps/platform/scripts/eval-review-run.ts`), we enforce a strict **Three-Pillar Blind Clinician Protocol**:

1. **Type-Level Information Barrier (`simulateBlindClinician`):** The function signature `simulateBlindClinician(complete, {manuscriptTitle, manuscriptMarkdown, finding})` physically omits `variant` (`clean`/`injected`), `injections.json`, `known-organic-errors.json`, and `split` (`dev`/`test`). The simulated corresponding author / biostatistician sees only the manuscript's Abstract, Methods, Markdown tables, local paragraph context, and the single review comment.

Table 2: Composition of the 19-Paper Full-Text Controlled Error-Injection Benchmark (`dev` $N = 11$, held-out `test` $N = 8$; 38 Full-Length Clean/Injected JATS XML Manuscripts, 57 Injected Objective Errors across E1–E7).

PMCID	Split	Study Design	Journal (2025) & Clinical Topic Summary	Injected Errors
PMC12644478	dev	rct	*Nat. Commun.* — Metformin on insulin resistance in adults with type 1 diabetes ($N = 40$)	E1, E3, E6
PMC12788160	dev	rct	*Nutrients* — Purified anthocyanins dose-response trial in middle-aged and elderly adults	E2, E5, E7
PMC12731986	dev	rct	*Curr. Oncol.* — Resistance exercise on QoL, anxiety, and depression in colorectal cancer	E1, E4, E5
PMC12685937	dev	retrospective_cohort	*Front. Psychiatry* — Antidepressants and hypertensive disorders in pregnancy cohort	E3, E6, E7
PMC12856494	dev	retrospective_cohort	*Front. Oncol.* — Risk factors for postoperative recurrence of desmoid tumors	E2, E4, E7
PMC12657621	dev	case_control	*Brain Behav.* — Vasomotor symptoms and ischemic stroke case–control study	E4, E5, E6
PMC12721543	dev	diagnostic_accuracy	*PLOS ONE* — TrueNat MTB plus assay diagnostic accuracy for pulmonary TB	E2, E3, E7
PMC12758107	dev	cross_sectional	*Glob. Heart* — Hypertension clinical epidemiology in rural Thailand cross-sectional survey	E1, E4, E6
PMC12876241	dev	meta_analysis	*Front. Med.* — PRISMA systematic review and meta-analysis of heat therapy for dysmenorrhea	E1, E3, E4
PMC12739847	dev	translational	*Gut Pathog.* — Lactobacillus probiotics in Blastocystis in vitro and in vivo models	E1, E5, E6
PMC12485006	dev	translational	*NPJ Sci. Food* — Mulberroside A capacity to inhibit senescence in vitro and in vivo	E1, E3, E5
PMC12839862	test	rct	*Curr. Oncol.* — Itraconazole reducing chemoresistance in advanced ovarian cancer RCT	E3, E5, E7
PMC12697981	test	rct	*PLOS Med.* — COMET-PCOS: Hormonal contraceptives and metformin in PCOS RCT	E1, E4, E6
PMC12681115	test	rct	*Cardiovasc. Diabetol.* — FIVE-STAR: Finerenone on arterial stiffness in T2D and CKD RCT	E2, E3, E5
PMC12118826	test	retrospective_cohort	*PLOS ONE* — Geriatric ocular trauma and mortality retrospective cohort	E2, E6, E7
PMC12654784	test	case_control	*Medicina* — Adipokines as prognostic biomarkers in multiple myeloma case–control	E1, E4, E7
PMC12774873	test	diagnostic_accuracy	*NPJ Prim. Care Respir. Med.* — Home spirometry diagnostic accuracy for asthma	E2, E3, E4
PMC12757420	test	cross_sectional	*Front. Immunol.* — Frailty and associated risk factors in Sjögren's disease survey	E1, E5, E6
PMC12703663	test	meta_analysis	*Alzheimers Dement.* — PRISMA meta-analysis of antipsychotic discontinuation in dementia	E1, E6, E7

2. **Strict Frozen Test-Split Boundary:** In cross-manuscript transfer (`dev` $N = 11 \rightarrow$ `test` $N = 8$), both blind clinician evolution (`evolveStoreFromFindingsBlind`) and missed-pattern checklist distillation (`evolveAuditChecklistsFromDevMisses`) execute *exclusively* on `dev`; attempting to invoke checklist distillation on `test` throws a hard runtime security exception, and the memory store is completely frozen throughout `test` evaluation.

3. **Post-Hoc Ground-Truth Calibration & Adversarial Profile:** Because Benchmark 1 possesses exact ground-truth labels (57 injected errors + 20 verified organic issues), every blind clinician decision (`accept_fix` vs. `rebut_false_positive`) is audited post-hoc against ground truth via a calibration matrix (True-Error Acceptance Rate, False-Rebuttal Rate, False-Positive Rebuttal Rate, Raw Digit Citation Rate, and Post-Guard Instance Leakage Rate). In addition, a hostile **Adversarial Author Profile** attempts to rebut *every* detected error across all 19 injected papers (`acceptedFix = false`) to stress-test `NON_EXEMPTABLE_SUBTYPES`.

Table 3: Clean-Room Benchmark Results across 19 Full-Length 2025 PMC JATS XML Papers (38 Clean/Injected Manuscripts, 57 Injected Objective Errors across E1–E7). Verified in-print organic errors caught on clean/injected papers (`unexpected_true`) are reported explicitly alongside true false positives.

Evaluation Regime & Split	Injected Recall (↑)	Organic True (Clean) (↑)	True FPs (Clean / Inj) (↓)	Precision (w/ Org) (↑)	F1 Score (↑)	Safety / Guard Verification
Clean-Room AST Only (`All`, $N = 19$)	**66.7**% (38/57)	**7**	**9** (5/4)	**85.5**%	**74.9**%	0 LLM Calls (0.26 Clean FP/p)
Clean-Room AST Only (`dev`, $N = 11$)	72.7% (24/33)	6	5 (3/2)	88.1%	79.7%	Raw Clean Alerts: 9 (6 Org + 3 FP)
Clean-Room AST Only (`test`, $N = 8$)	58.3% (14/24)	1	4 (2/2)	80.0%	67.4%	Raw Clean Alerts: 3 (1 Org + 2 FP)
Adversarial Rebuttal Stress (`All`, $N = 19$)	**66.7**% (38/57)	**7**	9 (5/4)	**85.5**%	**74.9**%	**0.0**% False Ex. (0/38 Dropped)
`pat_baseline` Pure-LLM (`All`, $N = 19$)	*[Configured in `run-injected-review-evolution.ts -llm`; awaiting `DEEPSEEK_API_KEY`]*					
`heurion_deep` Hybrid (`All`, $N = 19$)	*[Configured in `run-injected-review-evolution.ts -llm`; awaiting `DEEPSEEK_API_KEY`]*					
`heurion_evolved` Blind-Clinician (`test`)	*[Frozen `dev` → `test` transfer configured; awaiting `DEEPSEEK_API_KEY`]*					

Clean-Room Deterministic AST Per-Category Recall (38/57 = 66.7%, **Zero LLM Calls, Zero Study-Specific Regexes**):
E1 Sample Size: **8/10** (80.0%) E2 Arithmetic: **5/6** (83.3%) E3 Cross-Section/Table: **5/8** (62.5%) E4 95% CI Bounds: **6/8** (75.0%)
E5 Method–Result: **5/8** (62.5%) E6 Semantic Spin/Causal: **2/9** (22.2%) E7 Table Row/Cell Sum: **7/8** (87.5%)

5.3 Benchmark 2: Live `deepseek-chat` Evaluation on 7 Clinical Oncology Vignettes

To evaluate live LLM multi-stage reasoning and dual-track evolution on real clinical oncology error patterns, Benchmark 2 (`paper/benchmarks/run-7papers-review.ts`) executes live `deepseek-chat` (`temperature = 0`) across 7 curated clinical oncology manuscript excerpts (`SEVEN_PAPERS`, derived from published radiation oncology, target-trial emulation, thoracic imaging, and translational xenograft studies) containing 8 verified organic publication errors (including an inverted 95% CI [403, 1.688], uncleaned revision merge artifacts `60.062.3%`, a Table cell where $21/35 = 60.0\% \neq 84\%$, SD vs. SEM confusion, and an acronym definition conflict), comparing `zero_shot`, `pass_at_k` ($k = 3$), `pat_baseline` (4-stage text-only [6]), `heurion_deep`, and `heurion_evolved`.

6 Results and Analysis

6.1 Clean-Room AST Performance and Organic Error Discovery on 19 Full-Text PMC Papers

Table 3 and Table 4 report the verified Clean-Room Deterministic AST results (`-no-llm`, 0 LLM calls, 0 study-specific regexes) across all 38 full-length PMC manuscripts alongside the Adversarial Rebuttal Stress Test (with live `deepseek-chat -llm` hybrid rows marked pending API execution).

Architectural Division of Labor Confirmed on Full-Text Manuscripts. Table 3 reveals a crisp empirical division of labor across error categories when evaluated on unedited 6,000-word PMC JATS XML articles:

1. **High Zero-LLM Sensitivity on Mathematical & 2D Tabular Errors (E1, E2, E4, E7):** Without invoking any language model, the clean-room ProseMirror 2D table-grid and CI engine catches **87.5**% (7/8) of intra-table cell/row sum errors (**E7**), **83.3**% (5/6) of inline percentage errors (**E2**), **80.0**% (8/10) of sample-size flow contradictions (**E1**), and **75.0**% (6/8) of out-of-bounds 95% CIs (**E4**), with only 5 true false positives across all 19 full-length clean papers (0.26 per paper, Precision = **85.5**%).
2. **Why Semantic LLM Passes Are Necessary (E6 Spin & Subtle E3 Prose Conflicts):** Conversely, pure AST rules detect only 2/9 (22.2%) of **E6** spin/causal overclaims (where non-significant p-values reside in separate paragraphs or tables from the overclaimed conclusion, e.g., `PMC12644478-3` and `PMC12118826-1`) and 5/8 (62.5%) of **E3** prose-level conflicts (such as organ-list substitutions in `PMC12485006-3`). Capturing these open-ended

Table 4: Per-Study-Design Breakdown of the Clean-Room Deterministic AST Engine (`-no-llm`) and Adversarial Objective-Error Rebuttal Stress Test across the 19 Full-Text PMC Papers.

Study Design (N Papers)	**Injected Errors**	**Clean-Room AST Recall**	**Organic True (Clean)**	**True FPs (Clean / Inj)**	**Adversarial AST Recall**	**False Ex. Rate**
Randomized Controlled Trial ($N = 6$: 3 dev, 3 test)	18	12/18 (66.7%)	3	5 (3/2)	**12/18**	**0.0%**
Retrospective Cohort ($N = 3$: 2 dev, 1 test)	9	7/9 (77.8%)	0	0 (0/0)	**7/9**	**0.0%**
Case–Control Study ($N = 2$: 1 dev, 1 test)	6	4/6 (66.7%)	0	2 (1/1)	**4/6**	**0.0%**
Diagnostic Accuracy ($N = 2$: 1 dev, 1 test)	6	5/6 (83.3%)	4	0 (0/0)	**5/6**	**0.0%**
Cross-Sectional Survey ($N = 2$: 1 dev, 1 test)	6	3/6 (50.0%)	0	0 (0/0)	**3/6**	**0.0%**
PRISMA Meta-Analysis ($N = 2$: 1 dev, 1 test)	6	4/6 (66.7%)	0	2 (1/1)	**4/6**	**0.0%**
Preclinical / Translational ($N = 2$: 2 dev, 0 test)	6	3/6 (50.0%)	0	0 (0/0)	**3/6**	**0.0%**
Total / Overall ($N = 19$: 11 dev, 8 test)	**57**	**38/57 (66.7%)**	**7**	**9** (5/4)	**38/57**	**0.0%**

semantic contradictions without overfitting requires Stage 3 LLM reasoning—which in turn motivates Track A boundary memory to suppress LLM methodological false alarms.

Discovery of Genuine Unannotated Errors in Published 2025 Journals (`unexpected_true`). Across the 19 uncorrupted `clean` manuscripts, the clean-room AST engine emits only 12 alerts total—and manual verification against the published PDFs confirms that **7** of these 12 alerts (58.3%) are *genuine in-print publication errors* in peer-reviewed 2025 articles:

- `PMC12721543` **(*PLOS ONE*, 2025 — 4 CI Errors):** In both Table 3 (univariate cOR) and Table 4 (multivariate aOR), "Married/co-habiting" reports 0.34 (0.64–2.86) and "Divorced/widow" reports 0.15 (0.30–3.51), where point estimates lie entirely below their 95% confidence intervals.
- `PMC12644478` **(*Nature Communications*, 2025 — Table Percentage Error):** Table 3 "Maculopathy ($n, \%$)" in the Metformin arm ($N = 20$) reports "2 (20)" instead of 10%.
- `PMC12697981` **(*PLOS Medicine*, 2025 — Newly Discovered Table 4 Typo):** In Table 4 (Safety outcomes, Metformin $N = 81$), "Dysmenorrhea" reports "3 (4.7%)," whereas $3/81 = 3.70\%$ and five other 3-event rows in the exact same column correctly report "3 (3.7%)."
- `PMC12731986` **(*Current Oncology*, 2025 — CI vs. p-value Anomaly):** Table 2 "Fatigue" reports median difference -11.11 (95% CI -22.49 to 5.82, crossing zero) alongside $p = 0.040$, matching `known-issues.md`.

Meanwhile, the remaining 5 clean-paper alerts represent authentic clinical reporting boundaries—such as Statistical Methods sections declaring both parametric Mean ± SD and non-parametric Median [IQR] depending on Shapiro–Wilk normality (`PMC12731986`, `PMC12657621`), oncology RECIST `SD: stationary disease` vs. statistical `standard deviation` (`PMC12839862`), and meta-analysis primary-trial subgroup citations (`PMC12703663`).

6.2 Adversarial Rebuttal Stress Test and Zero-Leakage Guard Verification

As shown in the rightmost columns of Table 4, even when an adversarial user submits explicit same-sentence rebuttals (`acceptedFix = false`) attempting to dismiss all 38 detected objective errors across all 19 injected papers, `NON_EXEMPTABLE_SUBTYPES` blocks **100.0**% (38/38) of objective errors from being exempted (**0.0**% false-exemption rate). Furthermore, unit-test stress verification (`claims-pat.test.ts`) confirms that even when an LLM distiller intentionally emits raw sample sizes and percentages ($N = 240$, $n = 169$, 70.4%) in its first draft, `synthesizeSanitizedBoundaryRuleAsync` detects the leak (`firstDraftLeaked = true`), scrubs all numeric tokens, and passes `hasInstanceLeakage = false` (**0.0**% post-guard leakage).

Table 5: End-to-End Live `deepseek-chat` Deep Review Comparison across 7 Clinical Oncology Manuscript Vignettes (8 Verified Organic Publication Errors) against Google PAT Baselines [6].

Pipeline Mode	Organic Error Recall (↑)	False Positives Total (↓)	Precision (↑)	F1 Score (↑)	Active Boundary Rules (Leakage)
`zero_shot` (Single-Pass Full Text)	37.5% (3/8)	15	16.7%	23.1%	0 (0.0%)
`pass_at_k` ($k = 3$ Unfiltered Union)	62.5% (5/8)	26	16.1%	25.6%	0 (0.0%)
`pat_baseline` (4-Stage Text Only [6])	37.5% (3/8)	14	17.6%	24.0%	0 (0.0%)
`heurion_deep` (Cold-Start AST + 4-Pass LLM)	**100.0**% (8/8)	16	33.3%	50.0%	0 (0.0%)
`heurion_evolved` (Dual-Track Evolved)	**100.0**% (8/8)	**0** (−**100.0**%)	**100.0**%	**100.0**%	**4** (**0.0**%)

6.3 Live `deepseek-chat` Comparison Against Google PAT Baselines on 7 Oncology Vignettes

Table 5 presents the end-to-end live `deepseek-chat` comparison across the 7 clinical oncology manuscript vignettes containing 8 verified organic publication errors.

Single-pass `zero_shot` and text-only `pat_baseline` [6] detect only 3/8 (37.5%) of organic errors because they fail to cross-check 2D table cells against prose or catch typographical merge artifacts. Running $k = 3$ independent LLM passes (`pass_at_k`) raises recall to 62.5% (5/8) but inflates false positives to 26 (Precision = 16.1%). By contrast, `heurion_deep` detects **100.0**% (8/8) of organic errors, and `heurion_evolved` eliminates all 16 recurring clinical boundary false positives (such as Target Trial Emulation clone-censor-weight pseudocohort denominators and preclinical mouse-vs.-TMA cohort transitions), achieving **100.0**% precision and **100.0**% F1 with **0.0**% instance number leakage.

7 Related Work

Automated Scientific Peer Review and Claim Verification. Systems such as STORM [8] and PaperQA2 [7] automate literature synthesis, while FActScore [9] and ALCE [10] evaluate citation support in generated text. Recently, Jayaram et al. introduced Google's Paper Assistant Tool (PAT) [6], a multi-stage verification framework for scientific manuscripts. However, existing review pipelines operate over flat text and remain stateless across manuscripts. Our work complements multi-stage LLM review with a clean-room 2D ProseMirror table-grid scratchpad, precision-aware confidence interval verification, and a closed-loop CRDT comment self-evolution mechanism.

Self-Evolving LLM Agents and Memory Governance. Reflexion [11], MemGPT [12], and Voyager [13] pioneered verbal reinforcement and skill libraries for autonomous agents. Recent advances in self-supervised skill diffusion [14], commit-on-evidence memory (CoEM) [15], vantage-aware memory (SafeVantage) [16], modular skill evolution (MemSkill) [19], verifiable skill libraries (Skill-V) [17], evidence-grounded memory-to-skill co-evolution governance [18], and beyond-matched-experience transfer (MemTransfer) [20] have established rigorous foundations for agent memory governance. We bring these principles to clinical peer review, pairing zero-template Track A boundary distillation (`hasInstanceLeakage = false` and `NON_EXEMPTABLE_SUBTYPES`) with Track B procedural rule and audit-checklist crystallization.

8 Conclusion

We have presented **Heurion Deep Review**, a clean-room four-stage clinical manuscript verification architecture coupled with a comment-driven dual-track self-evolution loop (`evolveFromReviewCommentAsync`). Without relying on study-specific lexical heuristics or a single LLM call, our 2D ProseMirror table-grid AST engine detects 66.7% (38/57) of injected

objective errors across 19 full-length PMC manuscripts (80.0%–87.5% across sample-flow, arithmetic, and table-sum errors) at 85.5% precision (0.26 clean false positives per paper), uncovers 7 genuine in-print errors in published 2025 journals, and enforces a 0.0% false-exemption floor under adversarial rebuttal. Combined with quote-verified LLM passes and dual-track memory evolution on live `deepseek-chat` oncology vignettes, `heurion_evolved` achieves 100.0% recall (8/8) and 100.0% precision with 0.0% instance number leakage.

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